# GAME 10020

# ASSIGNMENT #3: AI SYSTEM IMPLEMENTATION

*BY JAKE KIELTY*

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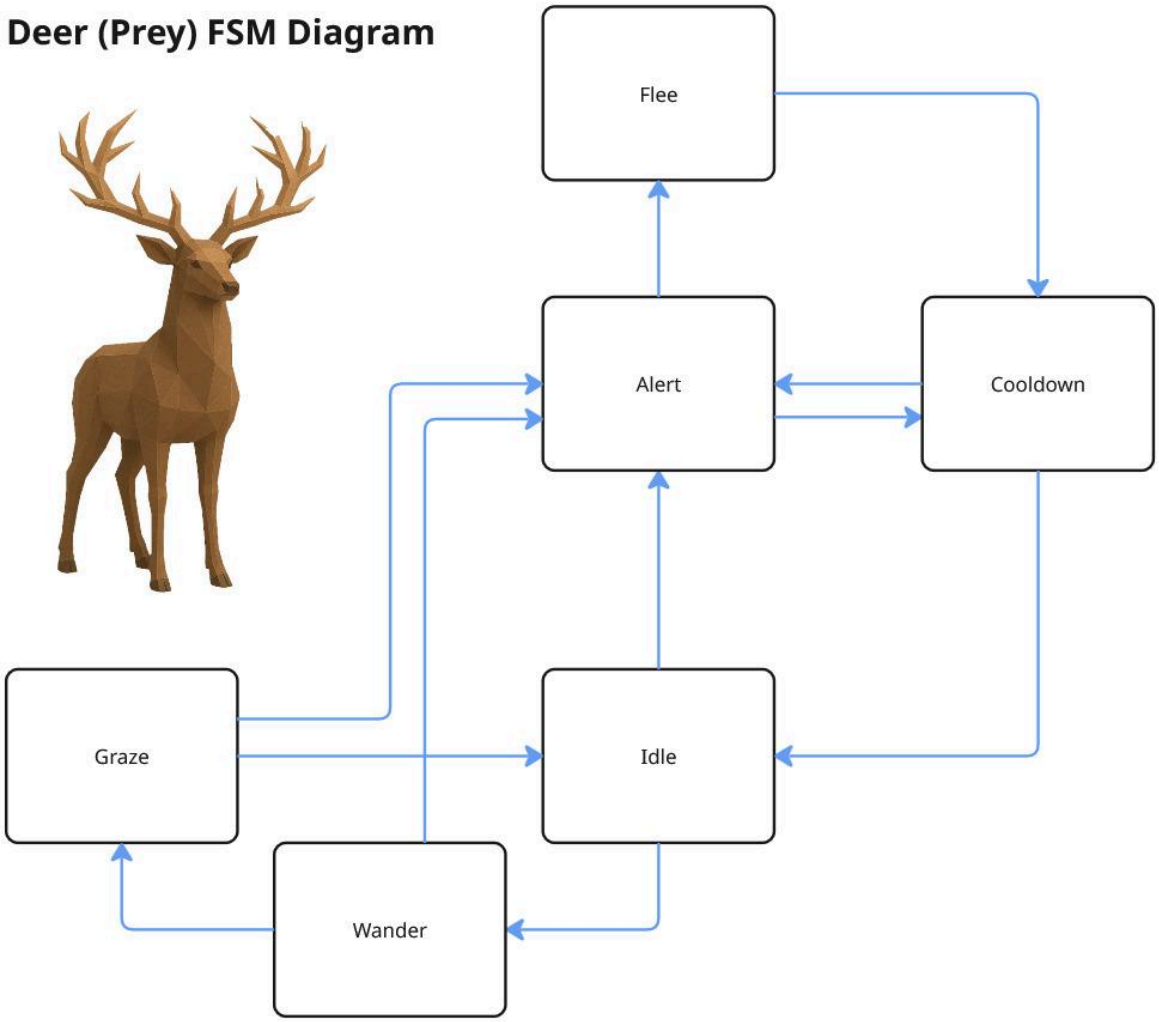
## FINITE STATE MACHINE DIAGRAM

This diagram is for a simple prey animal AI that roams around eating grass, and reacts to the player's presence by alerting when a player is detected, and running away when they get close. This AI generally uses vision as its primary sensor because it works well enough for what the AI is doing, and generally speaking, covers the bases of the other types of sensors well enough that I didn't need more to (roughly) emulate the (practically) near 360-degree sensory capabilities of a deer.

**Link to Video:** https://www.youtube.com/watch?v=mlvV89NNgqA



# BEHAVIORS PER STATE

## IDLE

This is the base “nothing” phase, where the AI will basically just stand there doing nothing. This exists as a connective phase between the graze/wander/return home states, and the alert, flee, and cooldown states. The AI will not remain here for too long, as it will be switching to other states often, but it will generally come back here before going to the others.

## ALERT

In the Alert state, the AI is aware that the player is there, and is now ready to take action if the player does something. In this case, while in the alert state, if the player moves close to the AI, it will enter the Flee state where it will run away. If the AI enters the Alert state, but does *not* enter the Flee state within a short period of time, it will go to the Cooldown state before going back to the Idle state. This is so that the AI won’t just immediately flee at first sight of the player. If the player remains detected, but doesn’t move closer, the AI will just stay in an Alerted state, basically waiting for the player. If the player Alerts the AI, but then exits detection, the AI will return to the cooldown state after a short period of time not detecting the player.

## FLEE

In the Flee state, the AI will attempt to move away from the player as much as possible with no regard for distance from home. This state only triggers when the AI has been Alerted, and then the player gets too close. This is different to the wander and graze states in that it has no regard for distance from home, as the creature is fleeing from a threat and doesn’t really care at the moment. This means it can end up far away from its home, but because return home can happen after cooldown, it will still end up returning home after the chase.

## WANDER

In the Wander state, the AI will periodically attempt to move around the area to random points in a defined area. This is basically just so that it doesn’t sit in one spot forever and never move. When in Idle, when nothing has happened for a short period of time, it will switch to Wander, where it will pick a spot, move to it, and then switch to Graze. When done with Graze, it will switch back to Idle, where it will repeat the process ad infinitum (or at least until it’s alerted). At any point, if it detects the player, it will be sent to the Alert state.

### GRAZE

In the Graze state, the AI will basically just stop what it's doing and play a basic animation for grazing on some grass. Once done with the animation, it will exit the Graze state into the Idle state, where it can repeat the Idle -> Wander -> Graze cycle over and over again, until interrupted by the player.

### COOLDOWN

This Cooldown state primarily exists to give the AI a way to distance itself after an Alert or Flee state. It also gives it more of an 'in-between' state where it's not fully alerted and but also not completely unaware. Once the AI has left the Alert or Flee states, it will have to go through this short cooldown before it can re-enter the Alert state. This also adds a bit of realism as it makes more sense to have these types of intermediate states than it does to only have either Detected or !Detected.

## SCENARIO

In this case, the scenario is fairly simple. You've got to herd the deer into the pen on the right side. For whatever reason, I have had some difficulties in getting the win-condition to work properly (collision has been annoying), so while getting the deer into the pen doesn't actually end the game, that's what the intention for this scenario is anyway.

## CHALLENGES & SOLUTIONS

During the course of working on this assignment, I have encountered several issues. For one, the biggest issue was that I haven't been able to wrap my head around the Polymorphic state machines, meaning I've been forced to stick with the simple state machines, which are more straightforward to work with.

A big issue also was that I wanted to originally go with the friendly combat robot ally design I made for the Game Systems Design course, but that would have been too complex with my current skills and with the amount of time remaining. As a compromise, I opted instead to go with the prey wildlife AI, as that would be a bit more straightforward (at least as far as I can tell) and would be more complex than the basic guard AI we went over.

One other issue that cropped up was that I found myself overcomplicating my states, to the point where they were starting to interfere with each other. I was finding that the Cooldown and Alert states were getting tangled and were basically firing off at the same time, and this ended up being caused by the conditions for the if statements being too all-encompassing, such that there were too many situations where both would be expected to fire, despite the fact that they were never intended to fire off together.

I also originally had a "Return Home" state, where the creature would move towards its starting area if it drifted too far away while fleeing, but this ended up being unnecessary as the creature already returns home because it is set to walk on a number of pre-set points. If I had the AI randomly pick where to go when Wandering/Grazing, then it'd need this return state, but as it stands, it doesn't need it, so I removed it.

Lastly, I wanted to have a basic win-state in here, so that you could see a victory screen after herding the deer into the pen, but unfortunately, I couldn't get that working in time. I've done it before on other projects, I just could not for the life of me get the collisions/triggers working for this this time.